

CPTG Finite-Polarization Jeans Closure of NGC 1052-DF2

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Abstract

Curvature Polarization Transport Gravity (CPTG) is tested here on the diffuse stellar body of the ultra-diffuse galaxy NGC 1052-DF2 using a pressure-supported radial Jeans reconstruction. A recent JWST tip-of-the-red-giant-branch measurement gives $D = 17.6 \pm 0.6$ Mpc. Rescaling the published photometric normalization to that distance gives $M_\star = 1.70368 \times 10^8 M_\odot$ and a documented major-axis effective radius $R_e = 1.92839$ kpc for the adopted Sérsic profile with $n = 0.6$. The current diffuse-light kinematic measurements are $\sigma_{\text{MUSE}} = 9.2^{+3.8}_{-4.5} \text{ km s}^{-1}$ and $\sigma_{\text{KCWI}} = 6.3^{+3.1}_{-3.7} \text{ km s}^{-1}$. Within the declared one-radius pressure-support reduction, the MUSE value supplies the structural-response anchor $a_\star = \sigma_{\text{MUSE}}^2 / R_e = 1.42243 \times 10^{-12} \text{ m s}^{-2}$. The projected Sérsic source is Abel-deprojected and normalized to the stellar mass, the nonlinear CPTG field is solved independently at each radius, and the complete radial field is then propagated through the Jeans equation and line-of-sight aperture projection. The isotropic stellar-only source gives $\sigma_{\text{ap}}(< R_e) = 7.22767 \text{ km s}^{-1}$, while the finite CPTG response gives $9.19500 \text{ km s}^{-1}$. At R_e , $g_{\text{CPTG}}/g_\star = 1.55102$ and the formal transport remnant is $1.46 \times 10^{-28} \text{ m s}^{-2}$, so the local finite response is overwhelmingly polarization-dominated. A centered $0.7R_e$ control gives $9.45769 \text{ km s}^{-1}$, approximately 1.02 upper-side quoted standard deviations above the KCWI central value. Because MUSE supplies the one-radius estimator of $a_\star$, the primary comparison is a shared-input closure rather than a held-out prediction. The retained solved field also gives projected $R_{95} = 4.29487$ kpc and Structural Mode $N = 1.56680$; under the current CSMI catalog this lies in the descriptive Magellanic-Irregular interval, with the continuous N value remaining the primary structural result rather than a visual morphology assignment. The calculation establishes a finite, low-response CPTG solution from the observed stellar source without inserting a dark-halo gravitational term; it does not claim a completed three-dimensional prolate CPTG field solution or that current dispersion uncertainties statistically require polarization over the stellar-only baseline.

1 Introduction: Why NGC 1052-DF2 Matters

NGC 1052-DF2 is an ultra-diffuse galaxy whose internal kinematics have been unusual since its discovery. The original analysis emphasized that its total dynamical mass appeared comparable to its stellar mass rather than to the much larger halo mass conventionally expected for a dwarf galaxy [1]. Subsequent diffuse-light spectroscopy with KCWI and MUSE provided direct stellar-body measurements, avoiding the need to infer the galaxy’s internal support solely from discrete globular-cluster tracers [2, 3].

The current observational picture has two features that are especially useful for a CPTG test. First, the stellar source is unusually diffuse and low-acceleration. Second, the directly measured stellar velocity dispersions are small enough that the gravitational contribution of the measured stellar mass itself is already important. Keim et al. recently revisited integrated-light broadening effects and report updated DF2 values of

$$\sigma_{\text{KCWI}} = 6.3^{+3.1}_{-3.7} \text{ km s}^{-1}, \quad \sigma_{\text{MUSE}} = 9.2^{+3.8}_{-4.5} \text{ km s}^{-1}. \quad (1)$$

These values remain consistent with a low dynamical mass and with the stellar source making a substantial fraction of the required gravitational support [4].

The distance scale has also changed. Tang et al. obtain a JWST tip-of-the-red-giant-branch distance

$$D = 17.6 \pm 0.6 \text{ Mpc}, \quad (2)$$

consistent with their independent surface-brightness-fluctuation result and smaller than the earlier HST TRGB estimate [5]. Because both physical radius and luminosity depend on distance, this measurement is adopted before the stellar source is reconstructed.

CPTG treats the observed baryonic system as the material source and computes a nonlinear curvature-polarization/transport response around it [10]. At the parent level, the theory organizes gravity through

$$G_{\mu\nu} + \Pi_{\mu\nu}^{\text{pol}} + \Theta_{\mu\nu}^{\text{trans}} = \frac{8\pi G}{c^4} T_{\mu\nu}, \quad (3)$$

where $T_{\mu\nu}$ is the measured material source, $\Pi_{\mu\nu}^{\text{pol}}$ is the curvature-polarization response, and $\Theta_{\mu\nu}^{\text{trans}}$ is the transport response. When the polarization and transport sectors vanish, the same material source reduces continuously to the Einstein equation. In the bound-system reduction used here, the observed stellar density first determines $g_\star(r)$; the object-dependent structural scale $a_\star$ then sets the strength of the finite CPTG response. Thus $a_\star$ is not a universal acceleration constant and the geometric response does not replace the baryonic source.

For a pressure-supported system, the field cannot be tested by identifying a measured velocity with a circular speed. The complete radial field must instead be propagated through the stellar Jeans equation and projected into the observed aperture. Structural Mode N is a separate post-solution law: it does not alter the field or fit the dispersion. It measures where radial change in the retained CPTG field is concentrated. Low N means that the dominant structural variation is weighted farther outward across the registered

radial interval, while progressively larger N means that the variation is organized farther inward. The CSMI name, when quoted, is assigned only after continuous N has been calculated and is therefore a descriptive catalog readout rather than a morphology input [12]. This source $\rightarrow$ field $\rightarrow$ Jeans $\rightarrow$ structural-readout ordering is the same methodology used in the CPTG pressure-supported validation of XMM-VID1-2075 [11].

The present calculation therefore asks a narrow question: can the observed diffuse stellar body of DF2 be carried through the CPTG source $\rightarrow$ radial field $\rightarrow$ Jeans projection chain while retaining a finite polarization response and without inserting a dark-halo gravitational source? The answer is tested first at the documented effective-radius scale and then through controlled source, anisotropy, distance, and aperture sensitivities. Once that radial field is retained, the calculation also evaluates the post-solution Structural Mode N , which measures where the solved radial response is organized and does not use a morphology label as an input [12].

2 Purpose, Test Definition, and Observed Inputs

The calculation is deliberately restricted to the diffuse stellar body. Globular clusters, planetary nebulae, and other discrete tracers are not used to set the primary source mass, distance, or pressure-support scale. The inputs have distinct roles: the distance fixes the physical scale and luminosity normalization; the Sérsic profile defines the stellar-source shape; the reference stellar mass-to-light ratio fixes the material-source normalization; MUSE supplies the declared one-radius pressure-support anchor; and KCWI is retained as a cross-instrument diffuse-light consistency control.

Operationally,

$$\begin{aligned} \{D, L_V, M_\star/L_V, R_e, n\} &\longrightarrow \rho_\star(r) \longrightarrow g_\star(r) \\ \sigma_{\text{MUSE}} &\longrightarrow a_\star \longrightarrow g_{\text{CPTG}}(r) \longrightarrow \begin{cases} \sigma_{\text{ap}}, \\ N. \end{cases} \end{aligned} \tag{4}$$

The measured dispersion and the theory response scale therefore enter different stages. The MUSE value sets the one-radius structural-response anchor under the declared reduction; it is not substituted directly for the final Jeans result. Structural Mode is later evaluated from the retained solved radial field and is not used to determine either $a_\star$ or the Jeans solution.

Observed quantity	Adopted value	Role in test
JWST TRGB distance	17.6 ± 0.6 Mpc	Physical scale and luminosity normalization
Major-axis effective radius	$R_e = 22.6''$	Primary radial and aperture scale
Projected Sérsic index	$n = 0.6$	Stellar-source profile shape
Projected axis ratio	$b/a = 0.85$	Geometry context and circularized sensitivity
V-band luminosity at 20 Mpc	$1.1 \times 10^8 L_\odot$	Stellar luminosity normalization
Reference stellar $M_\star/L_V$	2.0	Material-source mass normalization
MUSE diffuse-light dispersion	$9.2^{+3.8}_{-4.5} \text{ km s}^{-1}$	One-radius $a_\star$ anchor and closure comparator
KCWI diffuse-light dispersion	$6.3^{+3.1}_{-3.7} \text{ km s}^{-1}$	Cross-instrument consistency control

Table 1: Observed and source-reported inputs used to define the DF2 diffuse-stellar test. The primary radial calculation retains the documented major-axis effective radius; the projected flattening is examined separately through an area-preserving circularized-radius sensitivity.

At $D = 17.6$ Mpc, the angular effective radius becomes

$$R_e = 22.6'' \frac{17.6 \times 10^6 \text{ pc}}{206265} = 1.928393 \text{ kpc}. \quad (5)$$

The V-band luminosity and stellar mass scale as D^2 at fixed observed flux. Thus

$$L_V(17.6) = 1.1 \times 10^8 \left(\frac{17.6}{20} \right)^2 L_\odot = 8.5184 \times 10^7 L_\odot, \quad (6)$$

and with the reference $M_\star/L_V = 2$,

$$M_\star = 1.70368 \times 10^8 M_\odot. \quad (7)$$

This rescaling is a source normalization, not a dynamical mass inference from the measured dispersion.

3 Stellar Source and Local Structural Acceleration Scale

The projected stellar profile is represented by

$$\Sigma_\star(R) \propto \exp \left[-b_n \left(\left(\frac{R}{R_e} \right)^{1/n} - 1 \right) \right], \quad (8)$$

with $n = 0.6$. The primary calculation is a same-semimajor spherical reduction: the documented major-axis R_e is retained as the radial scale rather than silently replaced by a circularized radius. The three-dimensional tracer density is obtained by Abel inversion,

$$\rho_\star(r) = -\frac{1}{\pi} \int_r^\infty \frac{d\Sigma_\star/dR}{\sqrt{R^2 - r^2}} dR, \quad (9)$$

and normalized so that

$$4\pi \int_0^\infty \rho_\star(r) r^2 dr = M_\star. \quad (10)$$

The material-source field is then

$$g_{\star}(r) = \frac{GM_{\star}(< r)}{r^2}. \quad (11)$$

The pressure-supported interface requires an object-dependent structural response scale. The parent CPTG framework distinguishes the observed support coordinate σ^2/R_e from a universal definition of $a_{\star}$; a one-radius reduced calculation may, however, declare it as the estimator for that object-specific scale [10, 11]. The present calculation therefore uses

$$a_{\star} \equiv \frac{\sigma_{\text{MUSE}}^2}{R_e} \quad (12)$$

within this reduction only. With $\sigma_{\text{MUSE}} = 9.2 \text{ km s}^{-1}$ and Eq. (5),

$$a_{\star} = 1.42243 \times 10^{-12} \text{ m s}^{-2}. \quad (13)$$

The corresponding curvature-transport scale is

$$R_c = \frac{48\pi c^2}{a_{\star}} = 3.08782 \times 10^5 \text{ Gpc}. \quad (14)$$

The extremely large value of R_c anticipates that the explicit geometric transport remnant will be very small on kiloparsec scales, but it does not remove the polarization response from the field equation.

4 Radial CPTG Field Response

CPTG retains the stellar field as the source and solves the nonlinear curvature-polarization/transport response around it. The radial field is solved independently at each radius from [10, 11]

$$g(r) = g_{\star}(r) + \frac{\sqrt{a_{\star}g_{\star}(r)} + g_{\text{geom}}(r)}{\left[1 + (g(r)/a_{\star})^{123/50}\right]^{11/50}}, \quad (15)$$

where

$$g_{\text{geom}}(r) = \frac{1}{2}a_{\star} \frac{(r/R_c)^{7/5}}{1 + (r/R_c)^{12/5}}, \quad R_c = \frac{48\pi c^2}{a_{\star}}. \quad (16)$$

At the adopted major-axis effective radius, the source reconstruction gives

$$g_{\star}(R_e) = 1.94605 \times 10^{-12} \text{ m s}^{-2}, \quad (17)$$

while the solved field is

$$g_{\text{CPTG}}(R_e) = 3.01835 \times 10^{-12} \text{ m s}^{-2}. \quad (18)$$

Thus

$$\frac{g_{\text{CPTG}}(R_e)}{g_{\star}(R_e)} = 1.55102. \quad (19)$$

The local field is about 55.1% larger than the stellar-source field alone. The explicit transport remnant at the same radius is only

$$g_{\text{geom}}(R_e) = 1.46 \times 10^{-28} \text{ m s}^{-2}, \quad (20)$$

so the finite local response is overwhelmingly polarization-dominated.

Quantity at R_e	Value	Interpretation
$a_\star$	$1.42243 \times 10^{-12} \text{ m s}^{-2}$	One-radius structural-response anchor
$g_\star$	$1.94605 \times 10^{-12} \text{ m s}^{-2}$	Stellar material-source field
g_{CPTG}	$3.01835 \times 10^{-12} \text{ m s}^{-2}$	Solved total radial field
$g_{\text{CPTG}}/g_\star$	1.55102	Finite field-response factor
g_{geom}	$1.46 \times 10^{-28} \text{ m s}^{-2}$	Explicit transport remnant

Table 2: Primary same-semimajor spherical field solution at the documented major-axis effective radius. The finite correction is polarization-dominated at this scale.

A local field ratio is not itself a stellar-dispersion prediction. The stellar orbits contributing to an aperture sample a range of radii over which $g_{\text{CPTG}}/g_\star$ varies. The complete solved field is therefore passed through the Jeans equation rather than used as a multiplicative rescaling of a one-radius velocity dispersion.

5 Direct Dispersion-Supported Jeans Reconstruction

For a steady spherical tracer,

$$\frac{d(\rho_\star \sigma_r^2)}{dr} + \frac{2\beta_J(r)}{r} \rho_\star \sigma_r^2 = -\rho_\star g(r), \quad (21)$$

where β_J is the stellar orbital-anisotropy parameter. Defining

$$F(r) = \exp \left[\int_r^\infty \frac{2\beta_J(s)}{s} ds \right], \quad (22)$$

the physical outer-boundary solution is

$$\sigma_r^2(r) = \frac{1}{\rho_\star(r)F(r)} \int_r^\infty \rho_\star(s)F(s)g(s) ds. \quad (23)$$

The projected line-of-sight second moment is [6, 7]

$$\Sigma_\star(R)\sigma_{\text{los}}^2(R) = 2 \int_R^\infty \left(1 - \beta_J(r) \frac{R^2}{r^2} \right) \frac{\rho_\star(r)\sigma_r^2(r)r dr}{\sqrt{r^2 - R^2}}, \quad (24)$$

and the luminosity-weighted aperture moment is

$$\sigma_{\text{ap}}^2(< R_a) = \frac{\int_0^{R_a} \Sigma_\star(R)\sigma_{\text{los}}^2(R) 2\pi R dR}{\int_0^{R_a} \Sigma_\star(R) 2\pi R dR}. \quad (25)$$

The neutral primary calculation uses $\beta_J = 0$. The same tracer density and stellar mass normalization are solved twice: once with $g(r) = g_\star(r)$ and once with $g(r) = g_{\text{CPTG}}(r)$. This isolates the effect of the CPTG response while leaving the material source unchanged.

Jeans case	σ_{ap} (km s ⁻¹)	Role
MUSE diffuse-light anchor	$9.2^{+3.8}_{-4.5}$	One-radius support anchor and closure comparator
Stellar only, $\beta_J = 0$	7.22767	Baryonic baseline
CPTG, $\beta_J = 0$	9.19500	Primary finite-response closure
Shared-input closure residual Δ_{cl}	-0.00500	$\sigma_{\text{CPTG}}(< R_e) - \sigma_{\text{MUSE}}$; not held-out
CPTG, $\beta_J = -0.30$	8.84463	Tangential-anisotropy sensitivity
CPTG, $\beta_J = 0.22$	9.50300	Radial-anisotropy sensitivity

Table 3: Direct Jeans reconstruction using the complete radial stellar-only and CPTG fields. The MUSE dispersion also sets the one-radius estimator of $a_\star$, so the primary CPTG row is a shared-input closure rather than a held-out prediction.

For the isotropic primary case,

$$\sigma_\star(< R_e) = 7.22767 \text{ km s}^{-1}, \quad (26)$$

while

$$\sigma_{\text{CPTG}}(< R_e) = 9.19500 \text{ km s}^{-1}. \quad (27)$$

The shared-input closure residual relative to the MUSE support anchor is defined as

$$\Delta_{\text{cl}} \equiv \sigma_{\text{CPTG}}(< R_e) - \sigma_{\text{MUSE}} = -0.00500 \text{ km s}^{-1}. \quad (28)$$

The corresponding fractional closure residual is

$$\delta_{\text{cl}} \equiv \frac{\Delta_{\text{cl}}}{\sigma_{\text{MUSE}}} = -5.44 \times 10^{-4} = -0.0544\%. \quad (29)$$

A formal normalization by the quoted lower MUSE uncertainty gives $z_{\text{cl,shared}} = -0.00111$, but this quantity is not a held-out z -score and is not used for model selection.

This numerical closure is not interpreted as an independent statistical prediction because the same MUSE measurement enters Eq. (12). The nontrivial step is that after one scalar support coordinate fixes the response scale, the independently specified Sérsic source, nonlinear radial CPTG field, Jeans equation, and aperture projection return a self-consistent supported state.

The stellar-only value is also inside the broad current MUSE and KCWI intervals. Current diffuse-light uncertainties therefore do not statistically require polarization over the baryonic baseline. The stronger statement supported here is that a finite, nonzero CPTG polarization response is dynamically self-consistent and remains compatible with the observations.

6 Cross-Instrument and Dynamical Sensitivities

The KCWI observation samples a different footprint from the MUSE effective-radius aperture. Danieli et al. used a rectangular KCWI field offset toward the southwest of the galaxy center and extending through roughly $0.7R_e$ of the stellar body [2]. The present radial evidence set therefore does not claim an exact KCWI-footprint prediction. Instead it retains a centered circular $0.7R_e$ aperture as a controlled cross-instrument scale comparison.

For the isotropic CPTG solution,

$$\sigma_{\text{CPTG}}(< 0.7R_e) = 9.45769 \text{ km s}^{-1}. \quad (30)$$

Relative to the current KCWI value $6.3_{-3.7}^{+3.1} \text{ km s}^{-1}$, the upper-side normalized residual is

$$\frac{9.45769 - 6.3}{3.1} = 1.019. \quad (31)$$

Thus the centered $0.7R_e$ control is approximately a 1.02σ cross-instrument consistency result. Because the real footprint is not centered and circular, this number is used only as a scalar control, not as a precise KCWI prediction.

For a common comparison against standard MOND prescriptions, the MOND field is constructed from the same stellar source and passed through the same isotropic Jeans and aperture projection. The isolated comparison uses the simple interpolation function $\mu(x) = x/(1+x)$ with $a_0 = 1.2 \times 10^{-10} \text{ m s}^{-2}$ [8],

$$g_{\text{MOND,iso}}(r) = \frac{1}{2} \left[g_{\star}(r) + \sqrt{g_{\star}^2(r) + 4a_0 g_{\star}(r)} \right]. \quad (32)$$

Because the MOND prediction for DF2 depends on its external field, the one-dimensional EFE ansatz of Famaey, McGaugh, and Milgrom is also evaluated [9],

$$(g + g_e) \mu\left(\frac{g + g_e}{a_0}\right) = g_{\star}(r) + g_e \mu\left(\frac{g_e}{a_0}\right). \quad (33)$$

Famaey et al. adopted a co-distance 20 Mpc geometry with a projected separation of approximately 80 kpc, a host circular speed of 210 km s^{-1} , and therefore $g_e \simeq 0.15a_0$. The current distance measurements change the geometric inference. Tang et al. give $D_{\text{DF2}} = 17.6 \pm 0.6 \text{ Mpc}$ and $D_{\text{NGC 1052}} = 21.3 \pm 1.7 \text{ Mpc}$ [5]. Their tabulated sky coordinates give an angular separation of 13.651 arcmin, so the central distances imply

$$D_{\text{sep,central}} = 3.701 \text{ Mpc}, \quad (34)$$

and, using the same $g_e = V^2/D$ host prescription,

$$\frac{g_{e,\text{central}}}{a_0} = 0.00322. \quad (35)$$

A conditional co-distance geometry at 17.6 Mpc instead has a projected separation of 69.89 kpc and

$$\frac{g_{e,\text{codist}}}{a_0} = 0.17041. \quad (36)$$

The two cases are retained as a geometry bracket rather than interpreted as a statistical confidence interval. They bracket only the NGC 1052 contribution under the one-dimensional ansatz; they do not replace a full environmental MOND Poisson solution. Famaey et al. explicitly note that when the internal and external fields are comparable, the one-dimensional expression is an approximation and the exact treatment requires a numerical Poisson solver [9].

Figure 1 compares all models through the aperture-integrated line-of-sight stellar dispersion used in this paper. CPTG is orange and the stellar-only baryonic/Newtonian solution is blue. The current-central-distance MOND+EFE curve is solid green, the conditional geometry bracket is green shading, and the isolated MOND limit is dashed green. The observed MUSE and KCWI diffuse-light measurements carry their published asymmetric uncertainties. MUSE remains a shared-input closure comparator because it sets a_* . KCWI remains a cross-instrument control rather than an exact footprint prediction.

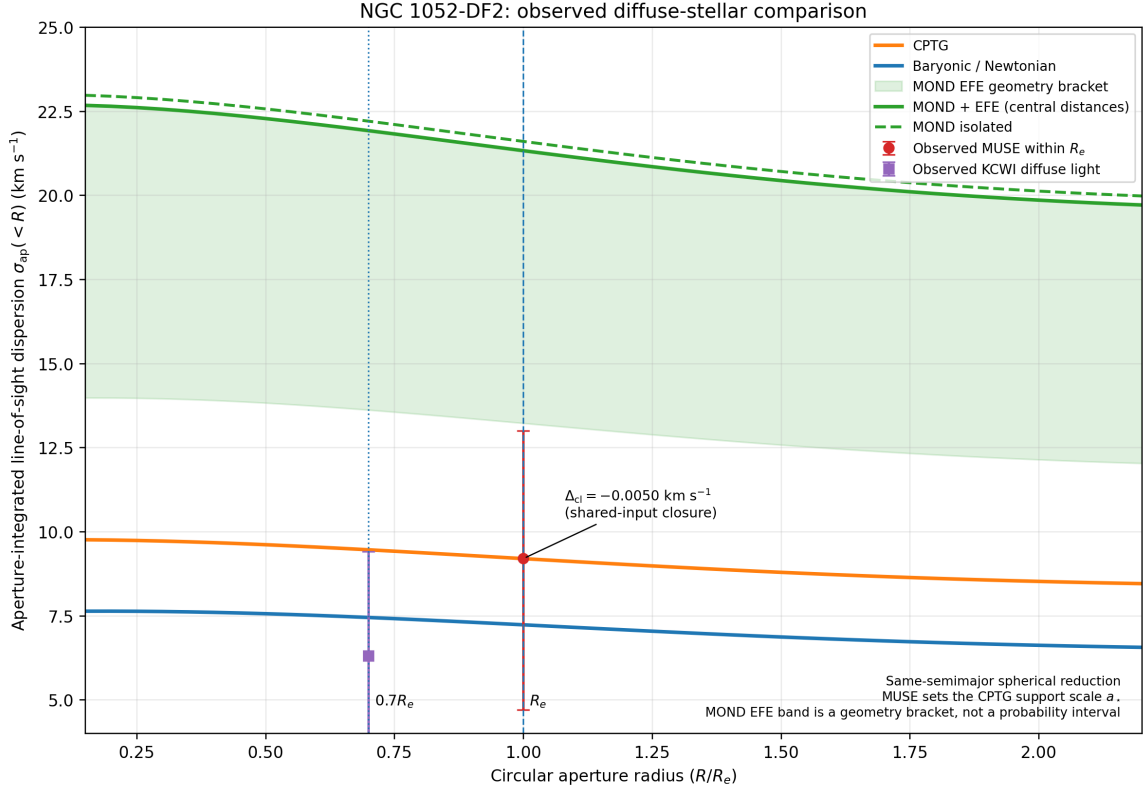


Figure 1: Observed diffuse-stellar comparison for the same-semimajor spherical reduction. The curves are aperture-integrated Jeans predictions from the same stellar source and projection machinery: CPTG (orange), baryonic/Newtonian (blue), MOND with the NGC 1052 EFE evaluated at the current central distances (solid green), and isolated MOND (dashed green). The green shaded region is a geometry bracket extending from the conditional co-distance strong-EFE case to the current-central-distance case; it is not a probability interval. The MUSE measurement at R_e anchors the CPTG support scale a_* and is therefore a shared-input closure point; the annotated Δ_{cl} is the corresponding closure residual, not a held-out fit residual. The KCWI diffuse-light measurement is shown at the nominal $0.7R_e$ scale as an observational control; its actual rectangular offset footprint is not reconstructed here. This is a pressure-supported dispersion comparison, not an observed rotation curve.

At the primary aperture, the conditional co-distance strong-EFE case gives

$$\sigma_{\text{MOND,EFE,codist}}(< R_e) = 13.222 \text{ km s}^{-1}, \quad (37)$$

while the current-central-distance geometry gives

$$\sigma_{\text{MOND,EFE,central}}(< R_e) = 21.324 \text{ km s}^{-1}. \quad (38)$$

The isolated limit is

$$\sigma_{\text{MOND,iso}}(< R_e) = 21.600 \text{ km s}^{-1}. \quad (39)$$

The original Famaey et al. nominal result $13.4^{+4.8}_{-3.7} \text{ km s}^{-1}$ corresponds to their close projected-separation assumptions [9]. Under the current central distance estimates, the NGC 1052 EFE becomes weak and the projected MOND dispersion approaches the isolated limit. The figure therefore reports the external-field geometry transparently rather than assigning one unconditional MOND prediction to DF2.

The documented $22.6''$ radius is a major-axis effective radius. The primary calculation intentionally preserves that semimajor scale. An area-preserving circularized radius provides a simple geometry sensitivity,

$$R_{e,\text{circ}} = R_e \sqrt{b/a} = 20.8362'' = 1.77789 \text{ kpc}. \quad (40)$$

Using that radius throughout the same spherical reduction gives

$$\sigma_{\text{CPTG}}(< R_{e,\text{circ}}) = 9.45852 \text{ km s}^{-1}, \quad (41)$$

so the basic finite-response closure is not created by the choice to retain the published semimajor radius.

Sensitivity	Range or alternate input	$\sigma_{\text{CPTG}}(< R_e) \text{ (km s}^{-1}\text{)}$
Orbital anisotropy	$-0.30 \leq \beta_J \leq 0.22$	8.8446–9.5030
Stellar M_*/L_V	1.5–2.5	8.3415–9.9454
JWST distance	17.0–18.2 Mpc	9.0857–9.3024
MUSE support anchor	$4.7\text{--}13.0 \text{ km s}^{-1}$	7.8785–10.3389
Radius convention	$R_{e,\text{circ}} = R_e \sqrt{0.85}$	9.4585

Table 4: One-at-a-time sensitivities of the finite-polarization Jeans solution. The radius-convention row uses the circularized radius consistently as the structural and aperture scale.

The widest controlled source-normalization sensitivity in this reduced calculation is the stellar mass-to-light ratio. The interval $1.5 \leq M_*/L_V \leq 2.5$ is a deliberately adopted sensitivity range rather than a source-measured uncertainty. Across that range, the resulting CPTG aperture values remain within the broad current diffuse-light dispersion range. The measured JWST distance uncertainty has a much smaller effect because the inferred stellar mass and physical radius change together.

7 Structural Mode N from the Solved DF2 Field

The Jeans calculation measures projected dynamical support. Structural Mode N asks a different question of the same retained CPTG solution: where is the radial change of the solved field concentrated? The mode is evaluated only after $g_{\text{CPTG}}(r)$ has been solved and retained; it is not a fit parameter and no visual morphology label is supplied to its calculation [12].

For a solved radial acceleration field, define

$$C_g(r) = g(r)^2 + \kappa_{\text{struct}}^2 \left(\frac{dg}{dr} \right)^2, \quad \kappa_{\text{struct}} = 1 \text{ kpc}, \quad (42)$$

and

$$w(r) = \left| \frac{dC_g}{dr} \right|. \quad (43)$$

Because an ideal Sérsic profile has no finite edge, the registered outer structural radius is the projected 95-percent-light radius,

$$\frac{\gamma(2n, b_n(R_{95}/R_e)^{1/n})}{\Gamma(2n)} = 0.95. \quad (44)$$

For $n = 0.6$ this gives

$$\frac{R_{95}}{R_e} = 2.227176979, \quad R_{95} = 4.294872689 \text{ kpc}. \quad (45)$$

The registered organization length and Structural Mode are then

$$\lambda = \frac{\int_{R_{95}/5}^{R_{95}} r w(r) dr}{\int_{R_{95}/5}^{R_{95}} w(r) dr}, \quad N = \frac{R_{95}}{\lambda}. \quad (46)$$

DF2 differs numerically from the monotone high-mode examples used in several previous CPTG structural calculations. Its $C_g(r)$ has one turning point on the registered interval, near $r = 1.9603$ kpc. The absolute weight in Eq. (46) is therefore integrated directly rather than replaced by a monotone endpoint identity. The primary same-semimajor spherical reduction gives

$$\lambda = 2.741171454 \text{ kpc}, \quad N = 1.566801917. \quad (47)$$

An independent higher-resolution calculation gives $N = 1.566801562$.

Structural case	N	Current CSMI readout
Primary same-semimajor sphere	1.56680	Magellanic Irregular
Area-preserving circularized radius	1.52290	Magellanic Irregular
Distance 17.0 Mpc	1.55007	Magellanic Irregular
Distance 18.2 Mpc	1.58171	Magellanic Irregular
$M_*/L_V = 1.5$	1.56841	Magellanic Irregular
$M_*/L_V = 2.5$	1.56430	Magellanic Irregular
MUSE-anchor lower endpoint, 4.7 km s^{-1}	1.53930	Magellanic Irregular
MUSE-anchor upper endpoint, 13.0 km s^{-1}	1.56825	Magellanic Irregular

Table 5: Structural Mode of the retained DF2 CPTG field and controlled one-at-a-time sensitivities. In the current CSMI map, $1.45 < N \leq 1.75$ carries the descriptive label Magellanic Irregular. The label classifies solved CPTG field organization and is not a visual morphology assignment.

The continuous result is therefore robustly low-mode under every sensitivity already carried by the evidence calculation. DF2 is observationally an ultra-diffuse galaxy, not a Magellanic irregular by visual classification. The CSMI name is retained only as the current catalog readout associated with the solved value $N \simeq 1.57$. The scientific result is the continuous mode itself: the dominant structural variation of the retained CPTG field is weighted relatively far outward compared with the high- N compact systems previously studied.

8 Results, Interpretation, and Limitations

The calculation supports five distinct statements.

First, the measured stellar light is sufficient to define the material source without reference to a dark-halo mass. At the adopted JWST distance, the source normalization is $M_* = 1.70368 \times 10^8 M_\odot$, and the stellar-only isotropic Jeans calculation gives $7.22767 \text{ km s}^{-1}$. This value is already compatible with the broad diffuse-light measurements. The baryonic source is therefore dynamically important before any CPTG response is added.

Second, the declared finite CPTG response remains nonzero. The MUSE support coordinate gives $a_* = 1.42243 \times 10^{-12} \text{ m s}^{-2}$, and the solved field at R_e is larger than the stellar field by a factor 1.55102. The explicit g_{geom} contribution is negligible on this local scale, so the additional field is overwhelmingly associated with the polarization term in Eq. (15). DF2 is therefore a low-acceleration, polarization-dominated local CPTG realization rather than an imposed zero-response endpoint.

Third, the radial solution closes through the pressure-supported observational interface. Passing the complete field through Jeans gives $9.19500 \text{ km s}^{-1}$ inside the same major-axis R_e used to define the support anchor. The cross-instrument $0.7R_e$ control is approximately 1.02σ above the KCWI central value on its quoted upper uncertainty. These are consistency results of the finite solution, not evidence that current data distinguish CPTG polarization from the stellar-only case.

Fourth, the retained radial field gives Structural Mode $N = 1.56680$ at the registered projected R_{95} . The result remains in the same $1.45 < N \leq 1.75$ CSMI interval under

every distance, stellar- M/L , support-anchor, and circularized-radius sensitivity evaluated here. The continuous low-mode value is the primary structural result; the associated Magellanic-Irregular CSMI name is a descriptive field-organization label and does not reclassify DF2’s observed morphology.

Fifth, no dark-halo source is inserted into the local field equation. This should not be confused with a numerical fit of a full conventional halo model. In the standard interpretation of DF2, the low measured dispersion is itself evidence that little or no dark matter is dynamically required [1, 2, 3, 4]. CPTG addresses the same bound-system state with different gravitational accounting: the material source remains the observed stellar mass, while the additional finite response is geometric polarization/transport rather than unseen matter.

Several limitations define the scope. The primary deprojection and Jeans projection are spherical even though DF2 is visibly flattened. The primary sphere retains the documented major-axis effective radius rather than attempting to encode the full intrinsic geometry. The circularized-radius calculation shows that this radial convention does not drive the main result, but it is not a replacement for an axisymmetric or prolate dynamical model. MUSE also reports a weak velocity gradient with prolate-like orientation [3]. That resolved ordered component is not solved by the present radial field calculation. A full three-dimensional CPTG prolate reduction would be required before the two-dimensional stellar velocity field could be claimed as a theory prediction.

The pressure-support scale is also shared with the primary comparator. The observed MUSE dispersion sets a_* , so the MUSE closure is intentionally not labeled a held-out prediction. KCWI provides a genuinely different instrument and aperture, but the present centered circular control does not reproduce its exact rectangular offset footprint. Finally, the reference stellar $M_*/L_V = 2$ is retained from the photometric source model rather than replaced by a new stellar-population fit at the updated distance.

These limitations make the result falsifiable rather than ambiguous. A stronger next test would retain the same stellar source and one solved CPTG response while fitting the actual two-dimensional MUSE stellar kinematic field and the exact KCWI footprint in an axisymmetric or prolate Jeans framework. Failure of one common field to support the resolved kinematics would reject that stronger DF2 reduction.

9 Conclusion

NGC 1052-DF2 provides a clean low-acceleration pressure-supported test because its diffuse stellar light, physical scale, and integrated stellar kinematics are all directly measured. Adopting the current JWST distance gives $R_e = 1.92839$ kpc and $M_\star = 1.70368 \times 10^8 M_\odot$ for the published Sérsic source and reference stellar mass-to-light ratio.

The current MUSE diffuse-light dispersion $9.2_{-4.5}^{+3.8} \text{ km s}^{-1}$ supplies the declared one-radius estimator $a_\star = 1.42243 \times 10^{-12} \text{ m s}^{-2}$. The stellar source is Abel-deprojected, the CPTG field is solved at every radius, and the complete field is passed through the Jeans and line-of-sight aperture equations. At R_e , $g_{\text{CPTG}}/g_\star = 1.55102$, while the explicit transport remnant is only $1.46 \times 10^{-28} \text{ m s}^{-2}$. The finite local correction is therefore polarization-dominated.

For the isotropic primary calculation, the stellar-only aperture dispersion is $7.22767 \text{ km s}^{-1}$ and the finite CPTG solution is $9.19500 \text{ km s}^{-1}$. The latter gives the explicitly defined shared-input closure residual $\Delta_{\text{cl}} = -0.00500 \text{ km s}^{-1}$ (-0.0544% of the MUSE central value). A centered $0.7R_e$ control gives $9.45769 \text{ km s}^{-1}$, approximately 1.02σ above the current KCWI central value on its quoted upper uncertainty. The result is stable under the current distance uncertainty, reasonable stellar mass-to-light variation, controlled orbital anisotropy, and an area-preserving circularized-radius sensitivity.

The retained solution also gives $R_{95} = 4.29487$ kpc, $\lambda = 2.74117$ kpc, and Structural Mode $N = 1.56680$. All controlled structural sensitivities remain in the same low-mode interval. Under the current CSMI catalog this maps to the descriptive Magellanic-Irregular interval, but the continuous N value—not the morphology name—is the physical post-solution result.

The scientific position is therefore narrow and explicit. DF2 admits a finite, low-response CPTG pressure-supported closure in which the observed stellar mass is the material source and curvature polarization modifies the radial gravitational field without adding a dark-halo gravitational term. The present diffuse-light uncertainties do not by themselves require polarization over the stellar-only baseline, so the result is a consistency closure rather than a model-selection detection. The next decisive extension is a common three-dimensional CPTG field tested against the resolved prolate-like MUSE kinematics and the exact KCWI observing footprint.

Data, Code, and Evidence Availability

The observational quantities used in this study are drawn from the published photometric and kinematic studies cited below. The primary distance is the JWST TRGB measurement of Tang et al. [5]. The stellar Sérsic structure, luminosity normalization, and reference mass-to-light ratio follow van Dokkum et al. [1]. The stellar-body spectroscopy is documented by Danieli et al. and Emsellem et al. [2, 3], with the current integrated-light dispersion values taken from Keim et al. [4].

The standalone numerical evidence package used for this paper is:

`CPTG_NGC1052-DF2_PUBLICATION_EVIDENCE_20260908.zip`

It contains the scalar inputs, radial Sérsic deprojection, enclosed stellar mass, stellar-only and CPTG acceleration profiles, projected Jeans aperture results, controlled sensitivities, the post-solution Structural Mode calculation and sensitivity table, the data and geometry bracket used for the observed CPTG/baryonic/MOND comparison figure, the shared-input closure-residual quantities, calculation code, independent higher-resolution verification, and cryptographic file hashes. The calculation is deterministic from the equations and inputs stated in the manuscript. Supporting CPTG theory and research materials are maintained at <https://github.com/CLG2025/CPTG>. Numerical implementations retain additional guard digits for reproducibility; manuscript values are rounded to the precision appropriate to interpretation.

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